Forecasting a Café's Daily Ingredient Demand with Damped Triple Exponential Smoothing

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***Abstract*—We present a lightweight forecasting pipeline that predicts the daily ingredient demand of a café from its raw sales history. Per-drink sales records are loaded into a relational database, joined with recipe data that is resolved into six base ingredients, and aggregated into one daily time series per ingredient. A self-implemented additive Holt–Winters model with a damped trend extrapolates each series into the future. In a two-month validation window, the model achieves a mean absolute percentage error of 74 % across all six ingredients, demonstrating that classical exponential smoothing remains effective for small-scale retail demand forecasting when ingredient quantities vary by orders of magnitude.**

***Index Terms*—forecasting, time series, triple exponential smoothing, Holt–Winters, damped trend, PostgreSQL**

I. Introduction

We present an end-to-end demonstration for a one-year sales log of a café (3 547 transactions, March 2024 to March 2025) provided as course material[1]. Sales and recipe data are integrated within a PostgreSQL database, aggregated into daily ingredient-usage series, and forecast by an additive Holt–Winters model with a damped trend, implemented from scratch. A command-line tool exposes every model parameter to the user, enabling interactive exploration of the forecast behavior.

II. Related Work

Demand forecasting in food service has long relied on classical time-series methods before the recent surge of deep-learning approaches. The Holt–Winters triple exponential smoothing method [2], [3] remains the standard baseline for data with both trend and seasonality, and its damped variant [4] addresses the common failure mode of linear trend extrapolation over long horizons. Subsequent work [5] formalized these methods within a state-space framework and established error metrics such as MAE and RMSE as universal evaluation criteria.

For small-scale operations without such auxiliary data, however, single-database pipelines with classical smoothing remain cost-effective. Our work shows that a self-contained PostgreSQL and Python stack with Holt–Winters provides a practical baseline for café inventory management.

[1]Datasets *Coffee sales* and *Coffee recipes* from the course *Databases and Information Systems*, Universität Hamburg, summer term 2026.

III. Architecture Overview

Figure 1 illustrates the data pipeline. The raw sales file `Coffee_sales.csv` is copied verbatim into the relational table `coffee_sales` in PostgreSQL; the CSV file is never accessed again, and all subsequent queries operate on the database alone.

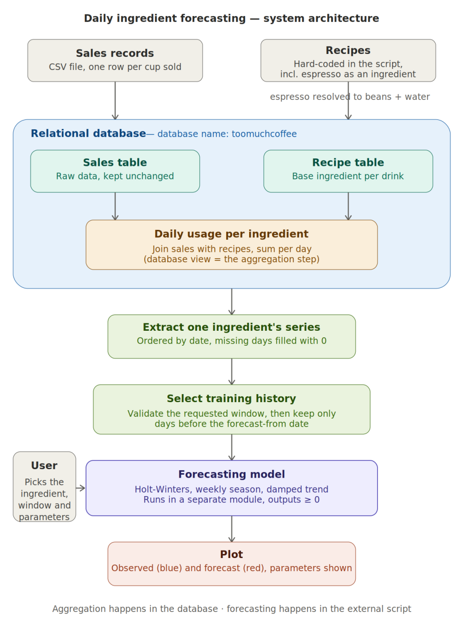

Fig. 1. Architecture of the demonstration: sales and recipe data are integrated in PostgreSQL, aggregated by a view, and consumed by the Python forecasting tool.

The recipes are derived from the completed JSON document collection of the second exercise sheet and embedded as a constant within the loader script. During loading, the pseudo-ingredient *espresso shot* is recursively resolved into its base ingredients (18 g coffee beans and 30 ml water per shot). A SQL view `daily_ingredient_usage` joins the recipe table with the sales table and aggregates per day and ingredient. The Python forecasting tool extracts one ingredient's time series from this view, fills days without sales with zero, forecasts the series, and renders the plot using `matplotlib`.

A. Forecasting Approach

We employ additive triple exponential smoothing (the Holt–Winters method) [1]–[3] with a seasonality period of $m = 7$ days, reflecting the pronounced weekly pattern in the data, extended by a damped trend [1], [4]. Given smoothing factors $\alpha, \beta, \gamma \in [0, 1]$ and a damping factor $\phi \in (0, 1]$, the level ℓ_t, trend b_t, and seasonal component s_t are updated for each observation x_t as follows:

$$\ell_t = \alpha(x_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1}), \tag{1}$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)\, \phi b_{t-1}, \tag{2}$$

$$s_t = \gamma(x_t - \ell_t) + (1 - \gamma)\, s_{t-m}, \tag{3}$$

and the h-step-ahead forecast is given by $\hat{x}_{T+h} = \ell_T + \left(\sum_{i=1}^{h} \phi^i\right) b_T + s_{T-m+1+((h-1) \bmod m)}$. Damping ($\phi < 1$) causes the trend to asymptotically approach a plateau rather than extrapolating linearly, which preserves the plausibility of long-horizon forecasts; setting $\phi = 1$ recovers the classical undamped model. The recursion is implemented in pure Python without reliance on any external forecasting library.

IV. Demonstration Proposal

The demonstration comprises an interactive command line session accompanied by live plots. Users first observe a preconfigured forecast and subsequently modify parameters to immediately observe the model's response.

A. Setup

The demonstration executes locally with PostgreSQL and Python 3.12, utilizing `psycopg2`, `argparse`, and `matplotlib`. A complete execution finishes in well under one second.

B. User Experience

Users interact with the tool through command line flags: `-i` selects among the six ingredients, `--forecast_from` and `--forecast_until` define the inclusive forecast window, and `-a`, `-b`, `-g`, `-p` control α, β, γ, and ϕ, respectively.

Two scenarios structure the walkthrough. The first, *validation*, positions the forecast window within the observed data range, such that the model sees only the history preceding the window; the resulting prediction can then be compared visually against actual values (Fig. 2), confirming that weekly seasonality is accurately captured. Quantitatively, over the two-month validation window, the model achieves a MAPE of 74% (spread: 60–112% per ingredient) and a MAE of 1.7–740 g per day. The high MAPE is driven by low-volume ingredients; MAE provides a more representative daily error estimate. The second scenario, *looking ahead*, positions the window beyond the final observed day (Fig. 3). Users are then invited to explore interactively – for instance, by switching to milk, setting $\phi = 1$ to observe the undamped trend drift across a long horizon, or setting $\alpha = 1$ to observe the forecast tracking noise rather than signal.

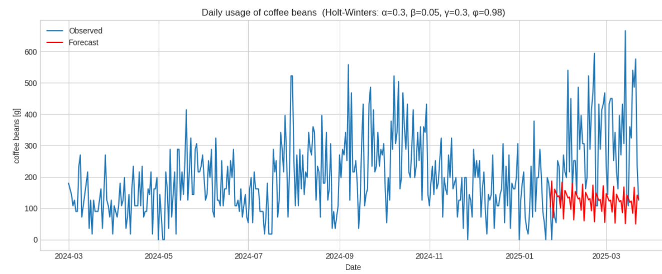

Fig. 2. Validation: forecast (red) of the last two observed months of coffee-bean usage, computed exclusively from the history preceding the window (α=0.3, β=0.05, γ=0.3, ϕ=0.98).

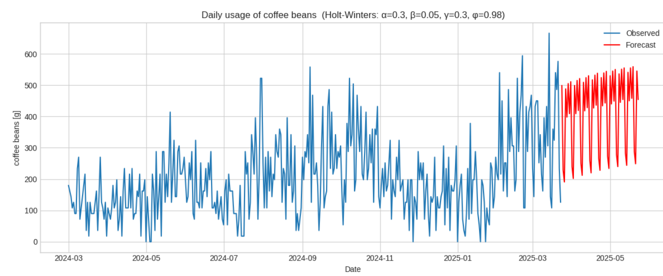

Fig. 3. Looking ahead: two-month forecast extending beyond the observed data, using the same parameters as in Fig. 2.

C. Conclusion

The demonstration illustrates that a complete ingredient-demand forecasting pipeline requires minimal infrastructure: a single relational database, one SQL view for drink-to-ingredient translation, and approximately one hundred lines of custom-implemented Holt–Winters code. Its distinguishing characteristic is full parameter transparency – every smoothing factor is exposed as a command-line argument, transforming the demonstration into an interactive platform for exploring exponential smoothing.

Limitations and Discussion

The demonstration shows that Holt–Winters smoothing provides a reasonable baseline for café ingredient forecasting, but several limitations apply. First, the model cannot account for one-off events such as holidays or special promotions [6]. Second, the deterministic recipe resolution does not capture substitution behavior, introducing systematic bias. Third, with only 3 547 transactions over one year, the data is too small for more complex models such as SARIMAX. Finally, the damping parameter ϕ was held fixed at 0.98; per-ingredient optimization could further reduce the MAPE.

Acknowledgment

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REFERENCES

[1] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021, ch. 8.3. [Online]. Available: https://otexts.com/fpp3/holt-winters.html

[2] C. C. Holt, "Forecasting seasonals and trends by exponentially weighted moving averages," *International Journal of Forecasting*, vol. 20, no. 1, pp. 5–10, 2004, reprint of ONR Research Memorandum 52, Carnegie Institute of Technology, 1957.

[3] P. R. Winters, "Forecasting sales by exponentially weighted moving averages," *Management Science*, vol. 6, no. 3, pp. 324–342, 1960.

[4] E. S. Gardner, Jr. and E. McKenzie, "Forecasting trends in time series," *Management Science*, vol. 31, no. 10, pp. 1237–1246, 1985.

[5] R. J. Hyndman, A. B. Koehler, J. K. Ord, and R. D. Snyder, *Forecasting with Exponential Smoothing: The State Space Approach*. Berlin, Germany: Springer, 2008.

[6] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.